

Using Protein Sequence Data to Test an Evolutionary Hypothesis

Did Aphids Acquire Their Ability to Make Carotenoids Through Horizontal Gene Transfer? Carotenoids are colored molecules that have diverse functions in many organisms, such as photosynthesis in plants and light detection in animals. Plants and many microorganisms can synthesize carotenoids from scratch, but animals generally cannot (they must obtain carotenoids from their diet). One exception is the pea aphid *Acyrtosiphon pisum*, a small, plant-dwelling insect whose genome includes a full set of genes for the enzymes needed to make carotenoids. Because other animals lack these genes, it is unlikely that aphids inherited them from a single-celled common ancestor shared with microorganisms and plants. So where did they come from? Evolutionary biologists hypothesized that an aphid ancestor acquired these genes by horizontal gene transfer from distantly related organisms.

How the Experiment Was Done Scientists obtained the DNA sequences for the carotenoid-biosynthesis genes from several species, including aphids, fungi, bacteria, and plants. A computer “translated” these sequences into amino acid sequences of the encoded polypeptides and aligned the amino acid sequences. This allowed the team to compare the corresponding polypeptides in the different organisms.

Data from the Experiment The sequences below show the first 60 amino acids of one polypeptide of the carotenoid-biosynthesis enzymes in the plant *Arabidopsis thaliana* (bottom) and the corresponding amino acids in five nonplant species,

using the one-letter abbreviations for the amino acids (see Figure 5.14). A dash (–) indicates a gap inserted in a sequence to optimize its alignment with the corresponding sequence in *Arabidopsis*.



INTERPRET THE DATA

- In the rows of data for the organisms being compared with the aphid, highlight the amino acids that are identical to the corresponding amino acids in the aphid.
- Which organism has the most amino acids in common with the aphid? Rank the partial polypeptides from the other four organisms in degree of similarity to that of the aphid.
- (a) Do these data support the hypothesis that aphids acquired the gene for this polypeptide by horizontal gene transfer? Why or why not? (b) If horizontal gene transfer did occur, what type of organism is likely to have been the source?
- What additional sequence data would support your hypothesis?
- How would you account for the similarities between the aphid sequence and the sequences for the bacteria and plant?

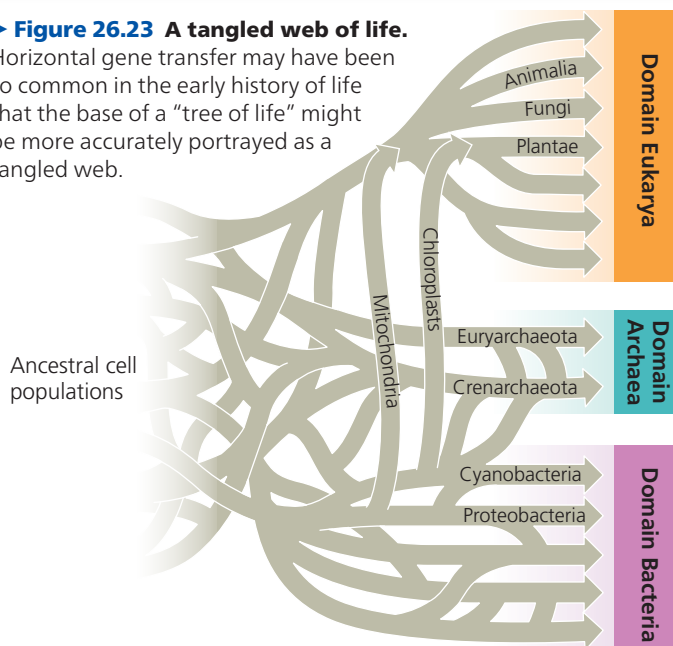
➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Data from Nancy A. Moran, Yale University. See N. A. Moran and T. Jarvik, Lateral transfer of genes from fungi underlies carotenoid production in aphids, *Science* 328:624–627 (2010).

Organism	Alignment of Amino Acid Sequences					
<i>Acyrtosiphon</i> (aphid)	IKIIIGSGV	GGTAAARLS	KKGFQVEVYE	KNSYNGGRCS	IIR-HNGHRF	DQGPSL--YL
<i>Ustilago</i> (fungus)	KKVVIIGAGA	GGTALAARLG	RRGYSVTVLE	KNSFGGGRCS	LIH-HDGRHW	DQGPSL--YL
<i>Gibberella</i> (fungus)	KSVIVIGAGV	GGVSTAARLA	KAGFKVTILE	KNDFTGGGRCS	LIH-NDGRHF	DQGPSL--LL
<i>Staphylococcus</i> (bacterium)	MKIAVIGAGV	TGLAAARIA	SQGHEVTIFE	KNNVVGGRMN	QLK-KDGFTF	DMGPTI--VM
<i>Pantoea</i> (bacterium)	KRTFVIGAGF	GGLALAIRLQ	AAGIATTVLE	QHDKPGGRAY	VWQ-DQGFTF	DAGPTV--IT
<i>Arabidopsis</i> (plant)	WDAVVIGGGH	NGLTAAAYLA	RGGLSVAVLE	RRHVIGGAAV	TEEIVPGFKF	SRCSTYLQGLL

► Figure 26.23 A tangled web of life.

Horizontal gene transfer may have been so common in the early history of life that the base of a “tree of life” might be more accurately portrayed as a tangled web.



connected branches (Figure 26.23). Although scientists continue to debate how best to portray the earliest steps in the history of life, in recent decades there have been many exciting discoveries about evolutionary events that occurred over time. We’ll explore such discoveries in the rest of this unit, beginning with Earth’s earliest inhabitants, the prokaryotes.

➔ **Mastering Biology**
Interview with Nancy Moran: Studying symbioses and horizontal gene transfer



CONCEPT CHECK 26.6

- Why is the kingdom Monera no longer considered a valid taxon?
- Explain why phylogenies based on different genes can yield different branching patterns for the tree of all life.
- MAKE CONNECTIONS** Explain how the origin of eukaryotes is thought to have represented a fusion of organisms, leading to extensive horizontal gene transfer. (See Figure 25.10.)

For suggested answers, see Appendix A.